

PPG Non-Invasive Glucometer

User Guide — Firmware v10

Sensor	MAX30102 (IR + Red LED)
Display	SSD1306 0.96" OLED 128x64
Controller	ESP32-C3 Mini
Battery	220mAh LiPo + TP4056 charger
Interface	Single push button
Measurements	Glucose · Heart Rate · SpO2 · Trend

This guide covers device operation, calibration, data logging, troubleshooting, and safety information for the PPG Non-Invasive Glucometer prototype. Read all sections before first use.

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1. Safety & Important Warnings

■ **This device is a research prototype. It is NOT a certified medical device and must NOT be used for medical diagnosis or treatment decisions.**

Not for medical use

Do not use readings to adjust medication, insulin doses, or make clinical decisions. Always consult a qualified healthcare professional.

Research only

This device is intended for research, academic, and educational purposes only.

Battery safety

Never short-circuit the LiPo battery. Keep away from heat and sharp objects. If the battery swells or leaks, stop use immediately.

Charging safety

Only charge using the TP4056 module. Never leave charging unattended for the first time. Red LED = charging, LED off = complete.

Sensor safety

Do not use if skin is broken, irritated, or infected at the measurement site. Remove immediately if the sensor feels hot.

Water

This device is not waterproof. Keep away from liquids.

2. Device Overview

The PPG Glucometer is a wrist-worn wearable that estimates blood glucose concentration non-invasively using photoplethysmography (PPG). Infrared and red light from the MAX30102 sensor pass through the fingertip and the reflected signal is analyzed to extract three physiological features that correlate with blood glucose.

2.1 Physical Components

OLED Display	0.96" 128x64 pixels, shows all readings and menus
Push Button	Single button on the right side — all interactions use Short, Long, and Double press
Power Switch	Slider switch on the top edge — ON/OFF
USB-C Port	Charging only (via TP4056 module)
Finger Clip	3D printed spring-loaded clip, MAX30102 sensor inside
LiPo Battery	220mAh, rechargeable, ~10-15 hours typical use

2.2 What It Measures

Blood Glucose	Estimated from IR peak, b/a ratio, and AC/DC ratio. Raw output corrected by personal calibration offset.
Heart Rate	Beats per minute, computed from inter-beat intervals during the 20-second scan.
SpO2	Blood oxygen saturation estimated from the Red/IR ratio using the R-ratio method.
Trend	Direction of glucose change over last 5 readings: ^ rising, v falling, - stable.

3. Getting Started — First Power On

3.1 First Time Setup

Before taking your first reading, charge the battery fully using a USB-C cable connected to the TP4056 charging port. The red LED on the charger module indicates charging in progress. When the LED turns off or changes colour, the battery is fully charged (approximately 30-45 minutes).

3.2 Powering On

1. Slide the power switch to the ON position (slider toward the right)
2. The splash screen appears for approximately 2.5 seconds
3. The device transitions automatically to the Idle screen
4. If this is the first use, the display shows "No reading yet"



Splash screen on boot



Idle — no prior reading



Idle — showing last result

3.3 First Scan Recommendation

For the most accurate first reading, take your initial scan in the morning before eating or drinking anything (fasting state). This establishes your personal baseline glucose level. After this, perform a calibration using a fingerstick glucometer reading for best accuracy.

■ *The first scan of any session reads slightly higher than subsequent scans due to thermal drift as the sensor warms up. Always take a throwaway scan first, then record the second scan onward.*

4. Button Gestures

The device uses a single push button for all interactions. Three gestures are available, each with a distinct timing:

Gesture		
SHORT press	< 500 ms	Confirm, scroll menu, increment digit, advance to next screen
LONG press	>= 1500 ms	Open menu, select menu item, abort scan, skip calibration
DOUBLE press	2 quick presses within 400 ms	Advance to next digit, confirm and exit

4.1 Screen Wake Gesture

When the OLED screen is off (auto-off after 30 seconds of inactivity), the first button press wakes the screen and redraws the current state. This first press is consumed — it does not trigger any action. A second press after the screen is on will perform the normal action.

■ *The screen never turns off during active scanning or finger detection — only during idle states, menus, and result screens.*

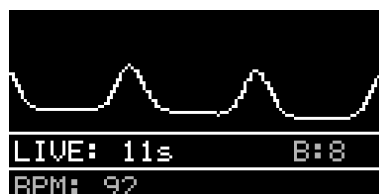
5. Taking a Glucose Reading

5.1 Step-by-Step Scan Procedure

From the Idle screen	Press SHORT to begin. The finger check screen appears.
Place finger in clip	Insert your index finger fully into the 3D printed clip. The spring should hold it with moderate pressure — not so hard that it hurts, not so loose that it wobbles.
Wait for detection	The device detects your finger automatically within 1-2 seconds. If it does not detect within 10 seconds, a timeout message appears and the device returns to idle.
Calibration phase (4s)	The first 4 seconds are silent calibration. The display shows CALIBRATING... — stay still during this phase.
Active scan (16s)	The waveform draws across the screen. Beat count and countdown timer update in real time. Stay completely still.
Results appear	After 20 seconds, results display automatically. If signal was too weak, a REJECTED screen appears with tips.



Waiting for finger



Scan in progress



Rejected — weak signal

5.2 Aborting a Scan

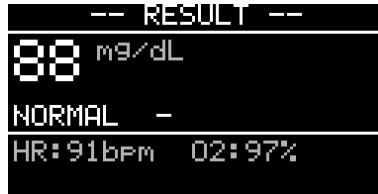
To cancel a scan at any time during the 20-second window, press and hold the button for 1.5 seconds. The ABORTED screen appears briefly and the device returns to idle. No reading is saved.

6. Understanding Your Results

6.1 Result Screen



Result — uncalibrated



Result — calibrated



Result — elevated with trend

The result screen shows four pieces of information:

Large number (top left)	Estimated glucose in mg/dL. Calibrated value if calibration has been performed, raw model output otherwise.
UNCAL label	Appears when no personal calibration has been applied. Results are less accurate until calibrated.
Classification + trend	Text classification (NORMAL, ELEVATED, HIGH, HYPO) and trend symbol (^ rising, - stable, v falling). Trend requires at least 3 prior readings.
HR and O2 (bottom)	Heart rate in BPM and SpO2 percentage measured during the same scan.

6.2 Glucose Classification

Label		
HYPO	< 70 mg/dL	Hypoglycaemic — low blood sugar
NORMAL	70–100 mg/dL	Normal fasting range
ELEVATED	100–126 mg/dL	Elevated — pre-diabetic range or post-meal
HIGH	> 126 mg/dL	High — diabetic range or post-meal peak

6.3 Trend Symbol

The trend symbol appears next to the classification after at least 3 readings have been taken. It reflects the direction of change across the last 5 readings using a median-smoothed linear regression, preventing single noisy scans from causing false trend changes.

^ (caret)	Rising — glucose increasing more than 2 mg/dL per reading on average
- (dash)	Stable — glucose change within ± 2 mg/dL per reading
v (v)	Falling — glucose decreasing more than 2 mg/dL per reading on average
(none)	Fewer than 3 readings available — trend not yet computed

6.4 After the Result Screen

The result screen dismisses automatically after 8 seconds, or immediately on a SHORT press. If this is your first reading (device is uncalibrated), the calibration prompt appears next. If already calibrated, the device returns to the Idle screen.

7. Calibration

Calibration corrects the systematic offset between the model's raw output and your personal true glucose level. Without calibration, readings may be consistently 10-25 mg/dL above or below your actual glucose.

■ Calibration data is stored permanently in the device memory (EEPROM) and survives power cycles. You only need to calibrate once initially, then add readings periodically to refine accuracy.

7.1 Calibration Procedure

```

CALIBRATE?
Device read:
102 mg/dL
Enter fingerstick
to calibrate now.
[5]YES [L]SKIP
  
```

Calibrate prompt

```

ENTER GLUCOSE
1 2 7
[5]+1 [H]scroll
[D]next [L]abort.
  
```

Enter fingerstick value

```

CONFIRM CAL?
Device: 102
True: 127
Offset: -25
[Short] SAVE
[Long] ABORT
  
```

Confirm before saving

Step 1 — Get a fingerstick reading: Use a standard blood glucose meter immediately before or after your device scan (within 2 minutes).

Step 2 — Calibrate prompt: After a scan, SHORT press to accept the calibration prompt, or LONG press to skip.

Step 3 — Enter the value: The digit entry screen shows 3 digits (hundreds, tens, units). SHORT press = +1, HOLD = fast scroll, DOUBLE press = advance to next digit.

Step 4 — Confirm: A confirmation screen shows the device reading, your entered true value, and the computed offset. SHORT press to save, LONG press to abort.

Saved confirmation: A SAVED! screen appears showing the offset and how many readings have been accumulated.

```

SAVED!
Offset: -25.0
Readings: 1
Recalibrate:
Menu > CALIBRATE
  
```

Calibration saved

```

CALIBRATION
Status: CALIBRATED
Offset: -25.0
Readings: 4
Weight: stable
Auto-return 4s
  
```

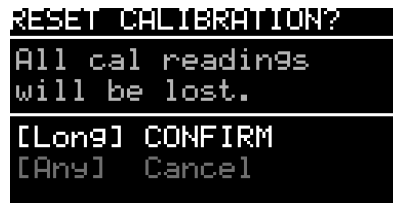
View calibration status

7.2 Calibration Strategy

- Take the first paired reading in the morning before eating (fasting baseline)
- Take a second paired reading 2 hours after a carb-heavy meal (post-meal peak)
- Add 1-2 more readings over the following week at different glucose levels
- After 4-5 readings the calibration is considered stable
- Do not calibrate more than once per day — variety matters more than frequency
- Always use the same finger as your normal scan finger

7.3 Resetting Calibration

To reset calibration (for example if you change finger or the readings have drifted), go to Menu → RESET CAL. A confirmation screen requires a second LONG press to prevent accidental deletion of your calibration data.



```
RESET CALIBRATION?  
All cal readings  
will be lost.  
[Long] CONFIRM  
[Any] Cancel
```

Reset cal — requires long press confirmation

8. Main Menu

Press and HOLD the button from the Idle screen to open the main menu. SHORT press scrolls through items. LONG press selects. DOUBLE press exits.



Main menu

CALIBRATE	Opens the calibration digit entry directly. Use this to add a paired reading from the menu without waiting for the post-scan prompt.
VIEW CAL	Shows current calibration status — offset value, number of accumulated readings, and weight stability. Displays for 4 seconds then returns.
RESET CAL	Clears all calibration data after double confirmation. Requires a LONG press on the confirmation screen to prevent accidents.
DATA LOG	Opens the data logging submenu for logging paired readings, exporting data to Serial, viewing record count, and clearing the log.
SET SUBJECT	Sets the subject ID (01-99) used to tag all subsequent data log records. Used for multi-subject research data collection.
EXIT	Returns to the Idle screen. DOUBLE press from anywhere in the menu also exits.

9. Data Logging

The data log stores paired scan results and true glucose values in the device's EEPROM memory. Up to 194 records can be stored. Each record includes IR peak, b/a ratio, AC/DC ratio, BPM, true glucose, and subject ID.

```
DATA LOG
>LOG READING
VIEW COUNT
EXPORT DATA
CLEAR DATA
< BACK
```

Data log submenu

```
SAVE LOG?
Glucose: 127 mg/dL
Subject: 01
Records: 12/194
[5] SAVE
[1] DISCARD
```

Confirm before saving

```
LOGGED!
Record #13 saved
Subject: 01
181 slots free
```

Record saved

9.1 Logging a Reading

1. Menu → DATA LOG → LOG READING
2. Finger check and 20-second scan run exactly as a normal reading
3. After the scan, a summary screen shows the measured features
4. Enter the exact fingerstick glucose value using digit entry
5. Confirm on the summary screen — the record is saved to EEPROM
6. The data log submenu appears again for immediate additional logging

9.2 Exporting Data

Connect the device to a PC via USB-C and open a Serial monitor at 115200 baud. Navigate to Menu → DATA LOG → EXPORT DATA. The device streams all records as CSV to the Serial port in this format:

```
DATA,subject_id,ir_peak,ba_ratio,ac_dc,bpm,true_glucose
```

Copy the Serial output and save as a .csv file. This file can be used directly with the `retrain_v10.py` script to retrain the glucose model with your personal data.

9.3 View and Clear

```
DATA LOG INFO
Records: 13
Free: 181
Max: 194
Subject: 01
Auto-return 4s
```

View record count

Menu → DATA LOG → VIEW COUNT shows total records stored, free slots, maximum capacity (194), and current subject ID.

Menu → DATA LOG → CLEAR DATA permanently deletes all records after a LONG press confirmation. Export your data before clearing.

10. OLED Auto-Off

To preserve battery life, the OLED display turns off automatically after 30 seconds of no button activity.

- The screen turns off cleanly — the device continues running normally
- Any button press wakes the screen and redraws the current state
- The first press that wakes the screen is consumed — it does not trigger an action
- A second press after waking performs the expected action normally
- The timer resets on every button press and every state transition
- The screen never auto-offs during scanning, finger detection, or the splash screen

■ *If you see a blank screen, press the button once to wake it. The device is still running and recording your scan even when the screen is off.*

11. Battery & Charging

11.1 Battery Life

Capacity	220 mAh LiPo
Active scan current	~120 mA
Idle current (screen off, sensor off)	~15 mA
Typical daily use (3-4 scans)	10-15 hours
Charging time	30-45 minutes (USB-C via TP4056)

11.2 Charging

- Slide the power switch to OFF before connecting USB for charging
- Connect USB-C to the TP4056 charging port
- Red LED on TP4056 = charging in progress
- LED turns off = battery fully charged
- Do not leave charging unattended for the first cycle
- The device can be used while charging with the switch in ON position

11.3 Battery Care

- Store with approximately 50% charge if not using for more than a week
- Do not store fully charged or fully discharged — this degrades LiPo cells
- Keep away from heat — do not leave in a hot car or in direct sunlight
- If the battery swells (puffs up), stop using immediately and dispose safely
- Charge nightly if using daily for continuous glucose monitoring sessions

12. Finger Placement Guide

Consistent, correct finger placement is the single most important factor for accurate and repeatable glucose readings. The 3D printed finger clip provides mechanical consistency — but correct initial positioning matters.

12.1 Correct Placement

- Always use the same finger — index finger of the dominant hand is recommended
- Insert the fingertip into the clip so the sensor window sits on the fleshy pad (the underside of the fingertip, not the nail side)
- The clip should hold the finger with moderate spring pressure — firm contact but not painful
- Let the clip close naturally — do not force or override the spring
- Keep the hand still and resting on a flat surface during the 20-second scan
- Relax — muscle tension in the hand reduces pulsatile blood flow and weakens the signal

12.2 Conditions That Affect Accuracy

Cold hands	Peripheral vasoconstriction reduces blood flow to fingertip. Warm hands briefly before scanning in cold environments.
Exercise	Wait at least 20 minutes after physical activity before scanning. Exercise redistributes blood flow away from periphery.
Caffeine	Strong coffee causes peripheral vasoconstriction, shifting the AC/DC ratio independent of glucose.
Thermal drift	The first scan of any session reads higher due to the cold sensor warming up. Discard the first scan of each session.
Wet fingers	Moisture at the sensor contact changes optical properties. Dry the finger completely before inserting into the clip.
Nail polish	Dark nail polish on the scan finger can absorb IR light and reduce signal strength. Use an unpainted finger.

12.3 Signal Quality Indicators

Use these real-time indicators during scanning to assess placement quality:

Beat count	Aim for 18-26 accepted beats per scan. Fewer than 10 usually leads to rejection.
Waveform	A smooth sine-like wave indicates good placement. Jagged, irregular waves indicate movement or poor contact.
SpO2 on result	95-100% confirms good signal. Values below 90% consistently indicate placement problems.
BPM on result	Should match your resting heart rate ± 10 BPM. Wildly wrong BPM indicates artifacts during scan.

13. Troubleshooting

Problem: Screen is blank

- Press the button once — the screen may be in auto-off mode
- Check the slider switch is in the ON position
- Check battery charge level — connect USB-C to charge
- If screen stays blank after button press, try power cycling (slider off then on)

Problem: REJECTED — Signal too weak

- Reinsert finger more firmly into the clip
- Make sure the sensor window is on the fleshy pad, not the nail side
- Warm hands if they are cold
- Try the other index finger
- Clean the sensor window with a dry cloth — dust reduces signal

Problem: Readings are inconsistent scan-to-scan

- Discard the first scan of each session (thermal warmup)
- Use exactly the same finger position every scan
- Wait 3 minutes between scans to allow thermal stabilisation
- Ensure the clip spring tension is consistent — check the hinge

Problem: Calibrated readings still seem wrong

- Reset calibration and start fresh with a new fasting baseline
- Ensure the fingerstick meter is calibrated and using fresh test strips
- Check you entered the fingerstick value correctly (3 digit entry)
- Add more paired readings — calibration stabilises after 4-5 readings

Problem: Finger not detected at finger check screen

- The IR threshold requires IR > 50,000 counts — press more firmly
- Wait for the 10-second timeout then retry
- Try a different finger or the other hand

Problem: SpO2 shows 80% or impossible values

- This usually happens on the first scan when the Red channel has not settled
- Discard the first scan and use the second onward
- Ensure the Red LED is making contact with the fingertip, not just the IR

Problem: Battery drains quickly

- Screen brightness and sensor scanning are the main consumers
- Let the screen auto-off between readings instead of keeping it on
- Charge nightly if doing multiple sessions per day

14. Technical Specifications

14.1 Hardware

Microcontroller	ESP32-C3 Mini (RISC-V, 160MHz, 400KB SRAM)
Sensor	MAX30102 (IR 880nm + Red 660nm dual LED)
Display	SSD1306 0.96" OLED, 128x64 pixels, I2C
Interface	Single tactile push button (GPIO10)
Power switch	Slider switch, routes LiPo+ through TP4056 to ESP32 5V pin
Battery	220mAh single-cell LiPo (3.7V nominal)
Charger	TP4056 with DW01A protection
Voltage regulation	AMS1117-3.3 onboard LDO (input 3.6-5.5V, output 3.3V 800mA)
I2C bus	SDA=GPIO8, SCL=GPIO9, 400kHz fast mode
Storage	4096 bytes EEPROM (calibration + data log)

14.2 Sensor Configuration

LED brightness	30 / 255
Sample average	1 (none)
LED mode	2 (IR + Red simultaneous)
Sample rate	100 Hz
Pulse width	411 us (maximum ADC resolution)
ADC range	4096 counts

14.3 Measurement Performance

Scan duration	20 seconds (4s stabilisation + 16s active)
Beat detection	Adaptive slope detector, 8-sample window, 400ms refractory
BPM range	40-190 BPM accepted
Glucose model	Polynomial degree-2 Ridge Regression (alpha=1000)
Features	IR peak amplitude, SDPPG b/a ratio, AC/DC ratio
Calibration	Weighted running average offset, converges in 4-5 readings
Repeatability (std)	~2.0 mg/dL scan-to-scan (with 3D printed clip)
Model MAE	~19 mg/dL pre-calibration (synthetic training data)
SpO2 method	R-ratio: $R = (AC_red/DC_red) / (AC_ir/DC_ir)$, $SpO2 = 110 - 25 \cdot R$

Trend history	5 readings, median-smoothed linear regression, threshold +/-2 mg/dL
Data log capacity	194 records (21 bytes each in 4KB EEPROM)
OLED auto-off	30 seconds inactivity timeout

15. Disclaimer

IMPORTANT DISCLAIMER

- This device is a RESEARCH PROTOTYPE developed for academic and educational purposes. It has not been evaluated, approved, or certified by any medical regulatory authority.
- This device is NOT intended for medical diagnosis, treatment, or management of any medical condition including diabetes mellitus.
- Glucose estimates produced by this device should NEVER be used to make medical decisions, adjust medication, determine insulin dosage, or replace clinical laboratory testing.
- The accuracy of this device is limited. Individual results vary significantly based on finger placement, skin characteristics, temperature, physiological state, and other factors.
- Always consult a qualified healthcare professional for medical advice regarding blood glucose management.
- The developers of this device accept no responsibility for any harm arising from the use or misuse of this prototype.

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